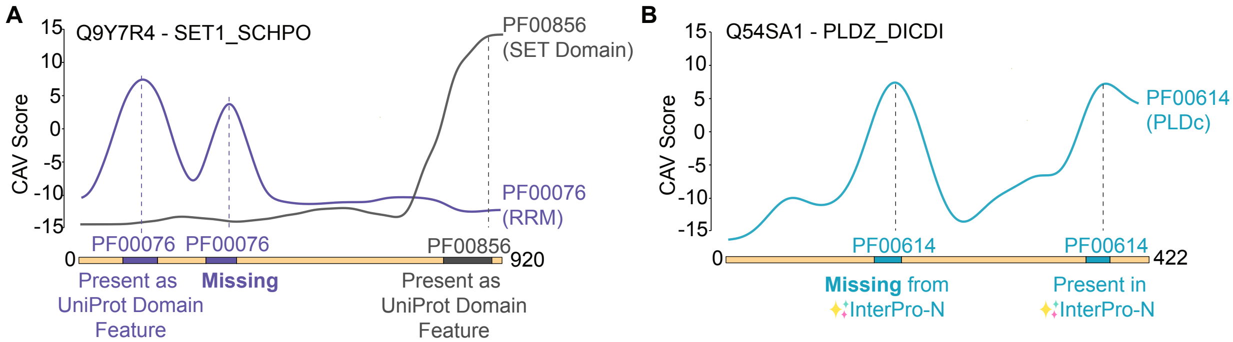

Figure 3: (A) CAV profiles can distinguish multiple occurrences of a motif or domain in a protein (A) CAV detects a peak for a second PF00076 domain in SET1_SCHPO, which is not annotated as a UniProt Domain Feature. (B) CAV detects a peak for a PLDc motif which is not detected by the InterPro-N deep learning model.

encode low-level patterns and late layers collapse information into task-specific summaries. This hierarchical pattern was first characterized in CNNs through feature-visualization studies [20], later generalized to deep neural representations and feature transferability [19].

Among these layers, layer 26 (indexed from 1, highlighted in Figure 2A) repeatedly appears as one of the highest-scoring and most stable across motifs. For this reason, all subsequent analyses and visualizations in this work use layer 26 as the default layer for computing alignment scores.

Example Localizations on Multi-Domain Proteins. Using layer 26, we illustrate the behavior of the CAV-based localization procedure on a representative protein, Q9NHV9/VAV_DROME, which contains four annotated domains for which concept vectors were trained. Figure 2B shows the smoothed CAV score profiles for each motif across the full sequence (just for layer 26).

Across all four motifs, the score trajectories exhibit the expected pattern: each motif attains a clear peak within its annotated ground-truth interval and remains low elsewhere. The transitions in the score profiles, both rising and falling, follow the boundaries of the annotated domains, indicating that the method is sensitive to motif locations rather than merely detecting coarse or diffuse signal. Qualitatively, a CAV score threshold of approximately five in this case appears to delineate motif-containing regions with high fidelity, capturing the annotated intervals while excluding adjacent non-motif residues.

Figures 3 demonstrates cases where the same domain or motif is present multiple times in the same proteins. These examples demonstrate that motif-specific concept vectors produce sharp, well-localized signals even in proteins with multiple occurrences of the same domain. Interestingly, the second peak of PF00076 RRM is missing from the current UniProt Domain Features for Q9Y7R4/SET1_SCHPO (Figure 3A). As we retrieved domain/motif coordinates from the UniProt API, this second peak initially appeared as a false positive in our evaluation. However, this second RRM is actually detected by several other detection methods and databases, including SMART Domains [12] and the InterPro-N model [3, 16]. Similarly, for Q54SA1/PLDZ_DICDI, while the first occurrence of PF00614 is clearly detected by a CAV peak, it is missed by InterPro-N (Figure 3B).

More broadly, these examples demonstrate the potential for CAVs to recover missed or weakly supported domain annotations at scale. Because they operate on residue-level embeddings rather than predefined sequence profiles, CAVs can highlight windows whose local representations resemble known motifs even when alignment-based methods offer limited signal. When applied systematically across proteomes, CAVs could offer a complementary signal to existing annotation systems and enable the continuous refinement of protein domain databases.

One consideration is that arrays of immediately adjacent occurrences of the same motif are sometimes localized by just one broader peak, with the array region generally covered by a CAV score of >= 5. More examples of CAV score profiles on proteins can be found in Appendix A.

Evaluation Across Motifs. To assess performance at scale, we constructed a benchmark consisting of ten randomly selected proteins per motif, yielding approximately 690 proteins in total. For each protein, we computed CAV scores for all 69 motifs and produced a ranked list of predicted windows. For evaluation, we adopted a simple but intuitive strategy: for each protein, we retained only the top-k scoring windows, where k equals the number of ground-truth motif instances.

We evaluated performance under varying overlap thresholds, summarized in Table 1. A predicted window is counted as a true positive if its overlap with a ground-truth motif exceeds a threshold defined as

$$\text{Overlap} = \frac{|\text{prediction} \cap \text{ground truth}|}{|\text{ground truth}|} \times 100.$$

The overall trend is consistent: performance remains strong at lenient to moderate thresholds, with F1-scores of 0.88, 0.87, and 0.87 at the 10%, 20%, and 30% overlap thresholds, respectively. Even at a 60% requirement, the method maintains a respectable F1-score of 0.80. Beyond this point, however, performance drops sharply, falling to 0.59 at 80% and 0.19 at 100%. This behavior indicates that the method very reliably identifies *where* motifs occur, but is less precise at reproducing their exact annotated boundaries.

Table 1: Motif detection performance across overlap thresholds.

Overlap	**Precision**	**Recall**	**F1**	**TP**	**FP**	**FN**
10%	0.9403	0.8289	0.8811	867	55	179
20%	0.9315	0.8317	0.8788	870	64	176
30%	0.9231	0.8260	0.8718	864	72	182
40%	0.9191	0.8260	0.8701	864	76	182
50%	0.8920	0.8136	0.8510	851	103	195
60%	0.8248	0.7830	0.8033	819	174	227
70%	0.7197	0.7046	0.7121	737	287	309
80%	0.5921	0.5899	0.5910	617	425	429
90%	0.4054	0.4054	0.4054	424	622	622
100%	0.1960	0.1960	0.1960	205	841	841

This limitation is not entirely surprising. Motif boundaries in curated databases are not always sharply defined, and many motifs have variable extents across homologs. Thus, the observation that predictions cluster strongly in the correct regions, yet do not always match the exact annotated endpoints, is consistent with the biological ambiguity inherent in motif definition.

Taken together, these results demonstrate that CAV-based scoring provides a reliable and biologically meaningful signal for localizing motif-rich regions across a diverse set of proteins. While the precise delineation of motif boundaries remains challenging, the method consistently identifies the correct regions with high confidence. These findings establish a strong foundation for more refined boundary-aware approaches in future work.

4. Conclusion

In this work, we show that functional protein motifs can be represented as coherent concepts within the embedding space of a pretrained protein language model. By training simple linear classifiers on motif and non-motif subsequences and extracting their normal vectors as Concept Activation Vectors, we find that motifs align with clear, interpretable directions that produce sharp localization signals across diverse protein families. These results indicate that many conserved structural patterns are embedded in PLM representations.

This perspective offers a compelling alternative to both alignment-based annotation and more complex end-to-end neural architectures. Unlike methods requiring persistent complex architectures, our entire detection system reduces to embeddings produced by pretrained PLM and a collection of motif vectors, one per concept. Once a motif-specific CAV is learned, detection reduces to a single inner product per window, making the approach highly interpretable, computationally lightweight, and readily extensible to large motif libraries. The consistency of our domain localization results suggests that PLMs encode structural regularities in a surprisingly structured and disentangled manner, with individual motifs occupying distinct, linearly separable regions of the representation space. Taken together, this framework provides a simple, scalable route for motif annotation that leverages information already present in modern PLMs.